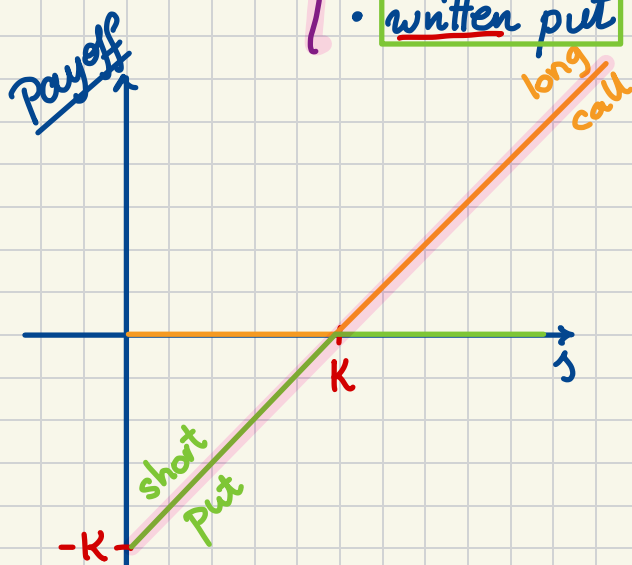


Put-Call Parity.

Portfolio A:

- long call
- written put

both **European**
& otherwise identical



The payoff f'n of the portfolio is $S - K$

The Payoff:

$$V_A(T) = V_C(T) - V_P(T) = \underline{S(T) - K}$$

Portfolio B:

- long non-dividend-paying stock
- borrow $PV_{0,T}(K)$ @ the risk-free rate to be repaid @ time T

$$\Rightarrow \boxed{V_B(T) = S(T) - K}$$

NO ARBITRAGE!

$$\Rightarrow V_A(0) = V_B(0)$$

$$\boxed{V_C(0) - V_P(0) = S(0) - PV_{0,T}(K)}$$

Put-Call Parity.

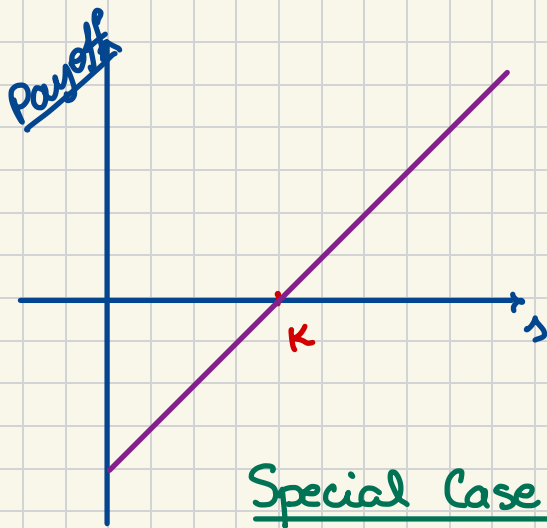
More generally:

for all $t \in [0, T]$:

$$\boxed{V_C(t) - V_P(t) = S(t) - PV_{t,T}(K)}$$

Remarks:

- The **no-arbitrage** assumption is sufficient.
- Only works for **European options**.



With Portfolio A,
we construct a
"synthetic" forward
or
"off-market" forward

Special Case:

strike price K = forward price for the stock F

$\Leftrightarrow$

$$K = F_{0,T}(S) = S(0)e^{rT} = FV_{0,T}(S(0))$$

$\Leftrightarrow$

$$PV_{0,T}(K) = S(0)$$

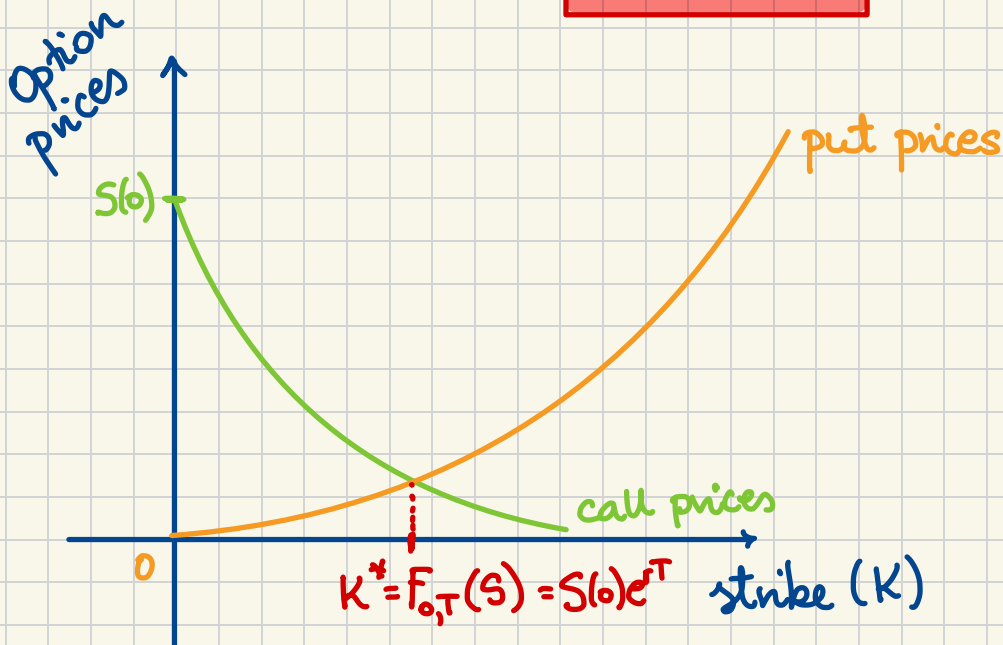
$\Leftrightarrow$

$$V_C(0) - V_P(0) = 0 = S(0) - PV_{0,T}(K)$$

$\Leftrightarrow$

$$\boxed{V_C(0) = V_P(0)}$$

Put-Call Parity.



Advanced Derivatives Questions

1. Consider a European call option and a European put option on a nondividend-paying stock. You are given:

- (i) The current price of the stock is 60.
- (ii) The call option currently sells for 0.15 more than the put option.
- (iii) Both the call option and put option will expire in 4 years.
- (iv) Both the call option and put option have a strike price of 70.

Calculate the continuously compounded risk-free interest rate.

$r = ?$

- (A) 0.039
- (B) 0.049
- (C) 0.059
- (D) 0.069
- (E) 0.079

Put-Call Parity.

$$V_C(0) - V_P(0) = S(0) - PV_{0,T}(K)$$

$$\underbrace{\quad}_{\text{ii (ii)}} \quad \quad \quad \downarrow$$

$$0.15 = 60 - 70e^{-4r}$$

$$70e^{-4r} = 59.85$$

$$\ln \mid e^{-4r} = \frac{59.85}{70}$$

$$-4r = \ln\left(\frac{59.85}{70}\right)$$

$$r = -\frac{1}{4} \ln\left(\frac{59.85}{70}\right) = \underline{0.03916}$$



77. You are given:

- i) The current price to buy one share of XYZ stock is 500
- ii) The stock does not pay dividends.
- iii) The continuously compounded risk-free interest rate is 6%.
- iv) A European call option on one share of XYZ stock with a strike price of K that expires in one year costs 66.59.
- v) A European put option on one share of XYZ stock with a strike price of K that expires in one year costs 18.64.

Using put-call parity, calculate the strike price, K .

- : $V_c(0) - V_p(0) = S(0) - PV_0(T)(K)$
- (A) 449
- (B) 452
- (C) 480
- (D) 559
- (E) 582

$$66.59 - 18.64 = 500 - Ke^{-0.06}$$

$$Ke^{-0.06} = 500 - 66.59 + 18.64$$

$$K = e^{0.06}(452.05) = 480.0032$$



78. The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. You are given that the price of a 35-strike call option is 3.35 higher than the price of a 40-strike call option, where both options expire in 3 months.

Calculate the amount by which the price of an otherwise equivalent 40-strike put option exceeds the price of an otherwise equivalent 35-strike put option.

- : $V_c(0, K_1=35) - V_p(0, K_1=35) = S(0) - 35e^{-0.08(0.25)}$
- (A) 1.55
- (B) 1.65
- (C) 1.75
- (D) 3.25
- (E) 3.35

$$\begin{aligned} & -V_c(0, K_2=40) - V_p(0, K_2=40) = S(0) - 40e^{-0.08} \\ & 3.35 - (V_p(0, K_1=35) - V_p(0, K_2=40)) = 5e^{-0.02} \\ & \text{answer} = 5e^{-0.02} - 3.35 = 1.55 \end{aligned}$$

